

STA 2023 - New Practice Final Exam

Spring 2026

Question 1

A transportation researcher at UCF wants to study whether a new carpooling initiative reduces students' average weekly commute time. The true average commute time of all UCF students is 22 minutes, but this value is unknown to the researcher. She selects 150 UCF students who volunteer to participate in the initiative. After 6 weeks, the average commute time of the sample decreases from 24 minutes to 18 minutes.

(a) Identify the following:

- i. Population
- ii. Sample
- iii. Parameter
- iv. Statistic

(b) Classify each of the following variables as **qualitative** or **quantitative**. If quantitative, state whether it is **discrete** or **continuous**.

- i. Weekly commute time (in minutes)
- ii. Primary mode of transportation (e.g., Car, Bus, Walk)
- iii. Number of days a student commutes to campus per week
- iv. Whether a student uses the carpooling initiative (Yes/No)

(c) Identify the level of measurement for each variable:

- i. Weekly commute time
- ii. Primary mode of transportation
- iii. Number of days a student commutes

(d) Is this study an **observational study** or a **designed experiment**? Justify your answer.

(e) Can the researcher conclude that the carpooling initiative **caused** the reduction in commute times? Explain clearly.

(f) Suppose instead the researcher randomly selected students from all UCF students using a student ID list.

- i. What type of sampling method is this?
- ii. Why is this method preferred over the original method used?

Question 2

A student organization records the weekly grocery spending, in dollars, for a group of 13 students: 40, 42, 45, 48, 50, 52, 55, 58, 62, 70, 85, 110, 145

Instruction: Show all steps. Answers without supporting work will not receive full credit.

(a) Find the five-number summary:

- i. Minimum
- ii. First quartile, Q_1
- iii. Median, Q_2
- iv. Third quartile, Q_3
- v. Maximum

(b) Construct a boxplot of the data. Clearly label:

- Q_1 , median, and Q_3
- Whiskers
- Any outliers

(c) Based on the boxplot, describe the shape of the distribution as symmetric, left-skewed, or right-skewed. Justify your answer.

Question 3

A university survey shows the following information about 250 students:

- 150 students subscribe to Netflix
- 100 students subscribe to Disney+
- 60 students subscribe to both Netflix and Disney+

A student is selected at random.

Instruction: Show all steps. Answers without supporting work may not receive full credit.

(a) Find the probability that the student:

- i. Subscribes to Netflix
- ii. Subscribes to Disney+
- iii. Subscribes to both Netflix and Disney+

(b) Find the probability that the student subscribes to **at least one** of the two services.

(c) Find the probability that a student subscribes to Disney+ **given that** the student subscribes to Netflix.

(d) Are the events “subscribes to Netflix” and “subscribes to Disney+” **independent**? Justify your answer using probability calculations.

(e) Suppose 3 students are selected at random **without replacement**. Find the probability that **all three** students subscribe to Netflix.

Question 4

A game is played where a player draws one card from a standard deck of 52 playing cards.

- If the player draws a Heart, they win \$5.
- If the player draws an Ace that is not a Heart, they win \$10.
- If the player draws any other card, they lose \$4.

Let X be the amount of money, in dollars, won in one play of the game.

Instruction: Show all steps. Answers without supporting work may not receive full credit.

- (a) Construct the probability distribution of the random variable X .
- (b) Find the expected value, $E(X)$.
- (c) Find the standard deviation of X .

Now suppose the game is played 12 times independently. Let Y be the number of times the player draws a Heart.

- (d) Find the probability that the player draws a Heart **at least 3 times**.
- (e) Find the mean and standard deviation of Y .

Question 5

The average weekly study time of students in a large university population is $\mu = 15$ hours with a standard deviation of $\sigma = 4$ hours. A random sample of size $n = 49$ students is selected.

Instruction: Show all steps. Answers without supporting work may not receive full credit.

- (a) Find the mean and standard deviation of the sampling distribution of the sample mean $\bar{X}$.
- (b) Find the probability that the sample mean study time is between 14 and 16 hours.

Now consider a different scenario: It is known that 40% of students at this university use AI tools for their coursework regularly. A random sample of size $n = 150$ students is selected. Let $\hat{p}$ be the sample proportion of students who use AI tools regularly.

- (c) Verify whether the normal approximation can be used for $\hat{p}$.
- (d) Find the mean and standard deviation of the sampling distribution of $\hat{p}$.
- (e) Find the probability that the sample proportion is greater than 0.45.

Question 6

A university library committee wants to estimate the proportion of students who support extending library hours until 2:00 AM on weekdays. A random sample of 300 students is selected, and 210 students say they support the idea.

Instruction: Show all steps.

- (a) Construct a 95% confidence interval for the true proportion of all students who support extending library hours.
- (b) Interpret the confidence interval in context.

Step-by-Step Solutions

Solution to Question 1

(a) Identifying Components

- i. **Population:** All UCF students.
- ii. **Sample:** The 150 UCF students who volunteered to participate in the initiative.
- iii. **Parameter:** The true average weekly commute time of all UCF students ($\mu = 22$ minutes).
- iv. **Statistic:** The average weekly commute time of the 150 students after the initiative ($\bar{x} = 18$ minutes).

(b) Classification of Variables

- i. **Weekly commute time:** Quantitative, Continuous.
- ii. **Primary mode of transportation:** Qualitative.
- iii. **Number of days a student commutes:** Quantitative, Discrete.
- iv. **Whether a student uses the initiative:** Qualitative.

(c) Levels of Measurement

- i. **Weekly commute time:** Ratio.
- ii. **Primary mode of transportation:** Nominal.
- iii. **Number of days a student commutes:** Ratio.

(d) Study Type

- **Answer: Observational Study.**
- **Justification:** The researcher did not randomly assign students to the carpooling initiative; instead, students volunteered (self-selection). The researcher is observing the outcomes of a group that chose to participate.

(e) Causation

- **Answer:** No.
- **Explanation:** Because this is an observational study with volunteers, there may be confounding variables (e.g., highly motivated students). Without a control group and random assignment, the researcher cannot establish a cause-and-effect relationship.

(f) Sampling Method

- i. **Method:** Simple Random Sampling.
- ii. **Why preferred:** It reduces selection bias (like volunteer bias) and ensures the sample is more representative of the entire population, allowing for better generalization of results.

Solution to Question 2

(a) **Five-Number Summary** Data set: 40, 42, 45, 48, 50, 52, 55, 58, 62, 70, 85, 110, 145 ($n = 13$).

i. **Minimum:** 40

ii. **First quartile, Q_1 :** Median of the lower half (40, 42, 45, 48, 50, 52). $Q_1 = \frac{45 + 48}{2} = 46.5$

iii. **Median, Q_2 :** The 7th item. 55

iv. **Third quartile, Q_3 :** Median of the upper half (58, 62, 70, 85, 110, 145). $Q_3 = \frac{70 + 85}{2} = 77.5$

v. **Maximum:** 145

(b) **Boxplot Construction**

Outlier Check

$$IQR = Q_3 - Q_1 = 77.5 - 46.5 = 31$$

$$1.5 \times IQR = 1.5 \times 31 = 46.5$$

$$\text{Upper Boundary} = Q_3 + 46.5 = 77.5 + 46.5 = 124$$

$$\text{Lower Boundary} = Q_1 - 46.5 = 46.5 - 46.5 = 0$$

Since $145 > 124$, 145 is an outlier.

- Box from 46.5 to 77.5 with a line at 55.
- Left whisker extends to 40.
- Right whisker extends to 110 (the largest value ≤ 124).
- An asterisk (*) at 145 represents the outlier.

(c) **Distribution Shape**

- **Answer: Right-Skewed.**
- **Justification:** The median (55) is closer to Q_1 (46.5) than to Q_3 (77.5), and there is a high-value outlier (145) on the right side, pulling the mean higher than the median.

Solution to Question 3

(a) **Basic Probabilities** Given: Total $N = 250$, $n(N) = 150$, $n(D) = 100$, $n(N \cap D) = 60$.

i. $P(N) = \frac{150}{250} = 0.6000$

ii. $P(D) = \frac{100}{250} = 0.4000$

iii. $P(N \cap D) = \frac{60}{250} = 0.2400$

(b) **Probability of At Least One**

Addition Rule

$$P(N \cup D) = P(N) + P(D) - P(N \cap D)$$

$$P(N \cup D) = 0.6000 + 0.4000 - 0.2400 = 0.7600$$

(c) **Conditional Probability**

Conditional Probability Formula

$$P(D|N) = \frac{P(N \cap D)}{P(N)}$$

$$P(D|N) = \frac{0.2400}{0.6000} = 0.4000$$

(d) **Independence**

- **Check:** Does $P(D|N) = P(D)$?
- **Comparison:** $0.4000 = 0.4000$.
- **Answer: Yes, the events are independent.** Because the occurrence of N does not change the probability of D occurring.

(e) **Multiplication Rule (Without Replacement)**

General Multiplication Rule

$$P(N_1 \cap N_2 \cap N_3) = \frac{150}{250} \times \frac{149}{249} \times \frac{148}{248}$$

$$P(N_1 \cap N_2 \cap N_3) \approx 0.6 \times 0.5984 \times 0.5968 \approx 0.2143$$

Solution to Question 4

- (a) **Probability Distribution of X** Given 52 cards: Hearts (13), Aces not Heart (3), Others (52 - 13 - 3 = 36).

- $P(X = 5) = \frac{13}{52} = 0.2500$
- $P(X = 10) = \frac{3}{52} \approx 0.0577$
- $P(X = -4) = \frac{36}{52} \approx 0.6923$

- (b) **Expected Value $E(X)$**

Expected Value Formula
$E(X) = \sum x \cdot P(x)$ $E(X) = 5(0.25) + 10(0.0577) - 4(0.6923)$ $E(X) = 1.25 + 0.577 - 2.7692 = -0.9422$

- (c) **Standard Deviation of X**

Variance and SD
$E(X^2) = 5^2(0.25) + 10^2(0.0577) + (-4)^2(0.6923) = 23.0968$ $\sigma^2 = E(X^2) - [E(X)]^2 = 23.0968 - (-0.9422)^2 = 22.2091$ $\sigma = \sqrt{22.2091} = 4.7127$

- (d) **Binomial Probability $P(Y \geq 3)$** Given: $n = 12, p = 0.25, q = 0.75$.

Complement Rule
$P(Y \geq 3) = 1 - [P(0) + P(1) + P(2)]$ $P(0) = (0.75)^{12} = 0.0317$ $P(1) = 12 \times (0.25)^1 \times (0.75)^{11} = 0.1267$ $P(2) = \binom{12}{2} \times (0.25)^2 \times (0.75)^{10} = 0.2323$ $P(Y \geq 3) = 1 - (0.0317 + 0.1267 + 0.2323) = 0.6093$

- (e) **Mean and SD of Y**

Binomial Mean and SD
$E(Y) = np = 12 \times 0.25 = 3.0$ $\sigma_Y = \sqrt{npq} = \sqrt{12 \times 0.25 \times 0.75} = 1.5$

Solution to Question 5

- (a) **Sampling Distribution of $\bar{X}$** Given: $\mu = 15, \sigma = 4, n = 49$.

Sampling Distribution of the Mean

$$\mu_{\bar{X}} = \mu = 15.0$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{49}} \approx 0.5714$$

- (b) **Probability $P(14 < \bar{X} < 16)$**

Z-score for Sample Mean

$$Z_1 = \frac{14 - 15}{0.5714} = -1.75, \quad Z_2 = \frac{16 - 15}{0.5714} = 1.75$$

$$P(-1.75 < \bar{Z} < 1.75) = P(Z < 1.75) - P(Z < -1.75)$$

$$0.9599 - 0.0401 = 0.9198$$

- (c) **Normal Approximation Verification**

Normality Conditions

$$np = 150 \times 0.40 = 60 \geq 10$$

$$n(1 - p) = 150 \times 0.60 = 90 \geq 10$$

Conclusion: Yes, the normal approximation can be used.

- (d) **Sampling Distribution of $\hat{p}$**

Sampling Distribution of the Proportion

$$\mu_{\hat{p}} = p = 0.40$$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.4 \times 0.6}{150}} = 0.04$$

- (e) **Probability $P(\hat{p} > 0.45)$**

Z-score for Sample Proportion

$$Z = \frac{0.45 - 0.40}{0.04} = 1.25$$

$$P(Z > 1.25) = 1 - P(Z < 1.25) = 1 - 0.8944 = 0.1056$$

Solution to Question 6

- (a) **Confidence Interval for p** Given: $n = 300, x = 210, \hat{p} = \frac{210}{300} = 0.70$.

Confidence Interval for a Population Proportion

$$\begin{aligned}\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \\ 0.70 \pm 1.96 \sqrt{\frac{0.7 \times 0.3}{300}} \\ 0.70 \pm 0.0519 \\ \text{Interval: } (0.6481, 0.7519)\end{aligned}$$

- (b) **Interpretation** We are 95% confident that the true proportion of all UCF students who support extending library hours until 2:00 AM is between 64.81% **and** 75.19%.